Granular Instrumental Variables

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We develop a new method to construct instruments in a broad class of economic environments. In the economies we study, a few large firms, industries, or countries account for an important share of economic activity. Idiosyncratic shocks to these large players significantly affect aggregate outcomes and are valid instruments. We provide a methodology to extract these idiosyncratic shocks to create granular instrumental variables (GIVs), which are size-weighted sums of idiosyncratic shocks. We show how GIVs enable us to estimate causal parameters, such as elasticities and multipliers, and are applicable in a broad range of empirically relevant settings.

I. Introduction

In many settings, there is a dearth of instruments, which hampers economists' understanding of causal relations (Ramey 2016; Stock and Watson 2016; Nakamura and Steinsson 2018; Chodorow-Reich 2019). We

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propose a new way to construct instruments. In the economies we study, idiosyncratic shocks that impact a few large actors, such as firms, industries, or countries, affect aggregate outcomes.[1] These idiosyncratic shocks (e.g., productivity shocks) are valid instruments for aggregate endogenous variables such as prices. We present a method to use these idiosyncratic shocks to construct granular instrumental variables (GIVs). The GIVs then allow us to estimate causal relations in a wide variety of economic contexts.

We first illustrate the idea in a basic static setup with supply and demand (sec. II). In this classic setting, we show how GIVs allow for a novel estimation procedure: they yield an instrument that allows us to estimate the price elasticities of both supply and demand. Intuitively, idiosyncratic demand shocks to large firms or countries provide valid instruments for changes in demand—and thus allow us to estimate the elasticity of supply. They also allow us to estimate the elasticity of demand: the idiosyncratic demand shock of a large firm impacts the price, which changes the demand of other firms. We formalize these ideas and present a way to extract and optimally aggregate idiosyncratic shocks, thus constructing optimal GIVs; the optimal GIV is the size-weighted sum of the idiosyncratic shocks. In this basic setup, we also show that some parameters that are of interest to economists, such as the pass-through of aggregate shocks to prices, can be estimated simply via ordinary least squares (OLS) using GIVs. We establish the consistency of the OLS and IV estimators using GIVs, and we derive their asymptotic distributions.

We analyze the instrument strength of GIVs in section III. The main insight is that GIVs are strong instruments when, first, a few large actors account for a substantial fraction of aggregate economic activity and, second, idiosyncratic shocks are volatile relative to the volatility of aggregate shocks.

In the same section, we also discuss the robustness of GIVs to various forms of misspecification. The IV estimator using GIVs is robust to including only a subset of the large actors, mismeasurement of the size distribution, and heterogeneity in demand elasticities. The OLS estimator using

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[1] Hence, economies are "granular"—their shocks are made of incompressible "grains" of economic volatility, the idiosyncratic shocks that occur at the level of firms, industries, or, in an international context, countries. This theme is laid out in Gabaix (2011) and developed in Acemoglu et al. (2012), di Giovanni and Levchenko (2012), di Giovanni, Levchenko, and Méjean (2014), and Carvalho and Grassi (2019).

GIVs does require that the size distribution is correctly measured. The main threat to identification when using GIVs is that we do not correctly isolate the idiosyncratic shocks. In this case, the estimated idiosyncratic shocks are affected by aggregate shocks, which can lead to bias. We provide several concrete solutions to mitigate that concern. First, we show that we can add factors for which we need to estimate the loadings. Second, we can "narratively check" GIVs: as the GIV procedure provides the dates and magnitudes of idiosyncratic and aggregate shocks, the largest idiosyncratic shocks can be analyzed in their historical context to confirm their idiosyncratic nature. Third, when a large idiosyncratic shock coincides with an unusual aggregate event (which we label a sporadic factor realization), it is prudent to remove those dates. Fourth, we can perform overidentification tests—for instance, split countries into two groups (e.g., developing vs. developed countries), form two GIVs using the idiosyncratic shocks of each group, and test whether the resulting estimates are statistically different.

Section IV shows how the basic procedure extends to a range of settings that are relevant to empirical research that include (but are not limited to) heterogeneous demand elasticities, heteroskedasticity, time-varying factor loadings, and multidimensional outcomes. This generality stems from the fact that the basic intuitive idea is general—namely, to use large, idiosyncratic shocks as primitive disturbances to the system.

Uses of GIVs.—Several recent papers have already applied GIV procedures to identify key parameters and elasticities of interest. Chodorow-Reich, Ghent, and Haddad (2021) study the multiplier of idiosyncratic shocks to an insurer's asset portfolio on the insurer's equity valuation. Camanho, Hau, and Rey (2022) study the impact of currency flows on exchange rates, using idiosyncratic shocks to fund-level rebalancing. Galaasen et al. (2021) use GIVs to study how idiosyncratic shocks to firms impact banks and how this spills over to other (small, nongranular) firms borrowing from the same bank. Schubert, Stansbury, and Taska (2022) study the impact of concentration on wages and use idiosyncratic firm-level shocks to instrument for concentration. Kundu and Vats (2021) estimate how idiosyncratic firm-level shocks in one state affect economic activity in other states via their transmission through the banking system. Adrian et al. (2022) use bank-level idiosyncratic shocks to estimate growth at risk. Ma, Paligorova, and Peydro (2022) estimate how lenders' expectations about a city affect GDP growth in the same geography. Dong, Kang, and Peress (2022) study the impact of flows on factor returns using GIVs. Flynn and Sastry (2022) use GIVs to study how narratives spread across firms over time. Baumeister and Hamilton (2023) use a GIV to estimate supply and demand elasticities and dynamics in the world oil market, using idiosyncratic supply shocks (e.g., Kuwait's production drop after it was invaded by Iraq in 1990) as granular instruments. We developed the GIV

while working on Gabaix and Koijen (2022), where we use it to measure the elasticity of demand for the aggregate stock market using idiosyncratic demand shocks to large investor sectors.

Related literature.—We relate to a number of strands. An active literature discusses identification strategies in macroeconomics (Ramey 2016; Nakamura and Steinsson 2018; Chodorow-Reich 2019; Huber 2023). We add to it by proposing GIVs, which provide a systematic candidate approach to identification.

A growing literature finds that a sizable amount of volatility is "granular" in nature—coming from idiosyncratic shocks to firms or industries (Long and Plosser 1983; Gabaix 2011; Acemoglu et al. 2012; di Giovanni and Levchenko 2012; di Giovanni, Levchenko, and Méjean 2014; Baqaee and Farhi 2019; Carvalho and Grassi 2019; Gaubert and Itskhoki 2021). We provide tools to isolate idiosyncratic shocks in the presence of common factors. Datasets used in this literature can be revisited, and GIVs can be constructed to investigate causal relations.

The idea of using idiosyncratic shocks as instruments to estimate spillover effects has been explored in several creative papers, as we discuss in more detail in appendix section D.1 (apps. B–D are available online), such as Leary and Roberts (2014), Amiti and Weinstein (2018), and Amiti, Itskhoki, and Konings (2019). However, the typical approach has been to use idiosyncratic shocks to variables that are excluded from the main estimating equation to construct instruments. We instead use the idiosyncratic shocks in the estimating equation directly. In addition, we allow for more flexible exposures to unobserved common shocks when extracting idiosyncratic shocks. Appendix section D.1 contains a fuller discussion of the literature.

Outline.—Section II introduces the GIV framework, centered around a classic model of supply and demand. Section III gives an economic discussion of the robustness and potential threats to identification and proposes diagnostic tests. Section IV presents a number of extensions. Section V concludes. Proofs are in appendix A or in the online appendix, which also presents additional extensions.

Notations.—For a vector $X = (X_i)_{i=1,\dots,N}$ and a series of weights w_i, we define $X_w = \Sigma_i w_i X_i$. With size weights S_i that satisfy $\Sigma_{i=1}^N S_i = 1$, we define $X_E := (1/N)\Sigma_{i=1}^N X_i$ and $X_S := \Sigma_{i=1}^N S_i X_i$ so that X_E represents the equal-weighted average of the vector's elements and X_S represents their size-weighted average. We also commonly use the notation u_i for shocks that are uncorrelated across i's and with variance $\sigma_{u_i}^2$. We use the notation θ_T^e for an estimator, based on observations from T periods, of a parameter θ whose true value is θ_0. We use C_t for a vector of controls; I for the identity matrix, $\iota = (1, \dots, 1)'$ for a vector of ones, all of the appropriate dimension given the context; and V^Y for the variance-covariance matrix of a random vector Y_t (so that $V^Y = \mathbb{E}[Y_t Y_t']$ if Y_t is a mean-zero column vector